

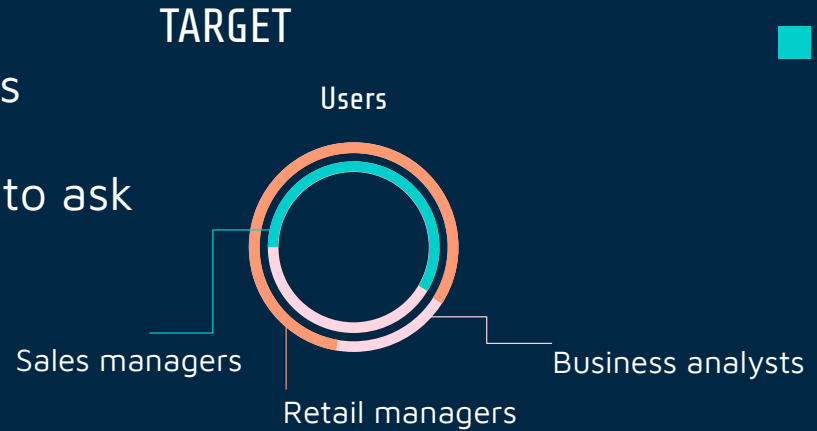
AI-Powered Retail Sales BI Agent

Final Exam Project
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Business Problem

Retail managers often need quick answers

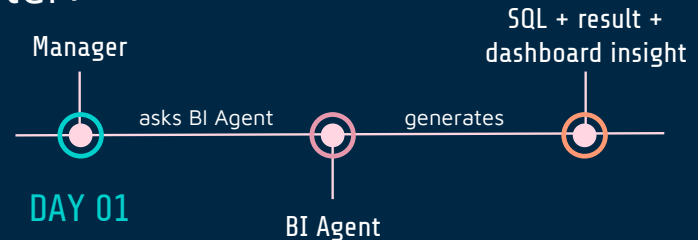
This project solves that by allowing users to ask business questions in natural language



Before:



After:



Dataset and Project Goal

- Dataset: Online Retail II from Kaggle

Raw column	Meaning
 InvoiceNo	Invoice identifier for each transaction
 StockCode	Product/item code
 Description	Product name or item description
 Quantity	Number of units sold
 InvoiceDate	Timestamp of the invoice
 UnitPrice	Price per unit
 CustomerID	Customer identifier
 Country	Customer country

Project Goal

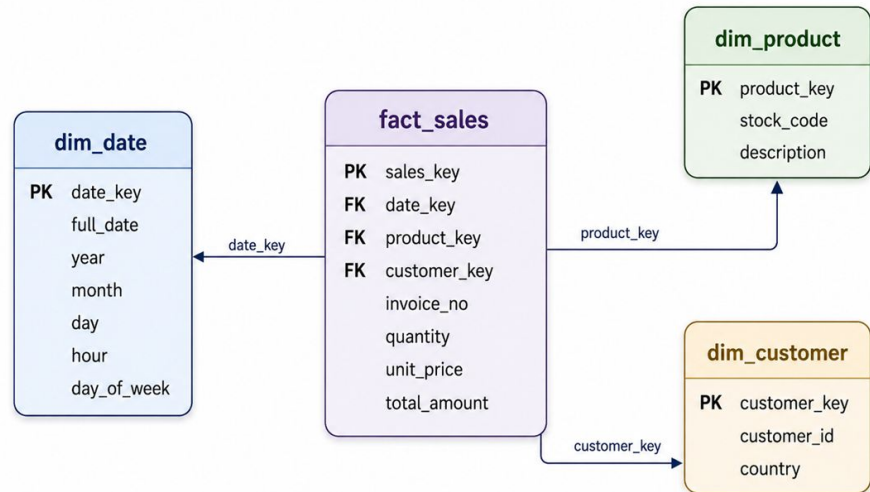


Build an
end-to-end BI
system

Star Schema Data Architecture

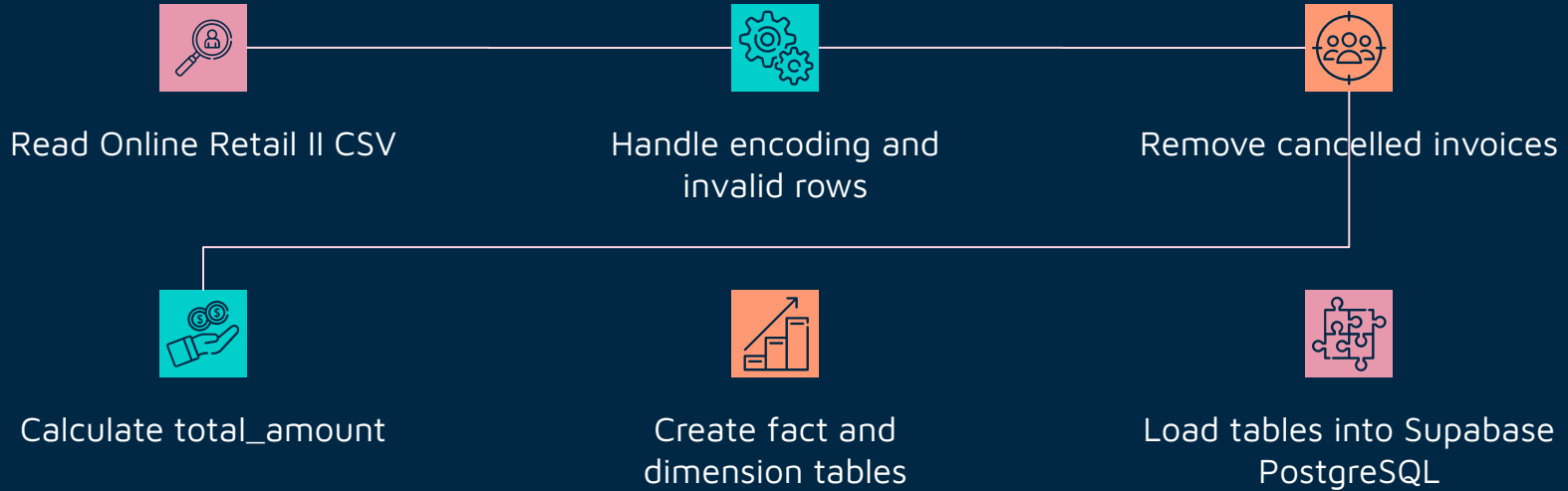
- The raw dataset was normalized into a star schema.
- The central table is fact_sales, which stores measurable sales transactions.
- Dimension tables store date, product, and customer information.

Retail Sales Star Schema ERD



PK = Primary Key, FK = Foreign Key

ETL Pipeline and Validation



dim_date = 17,282



dim_customer = 4,346



dim_product = 3,897



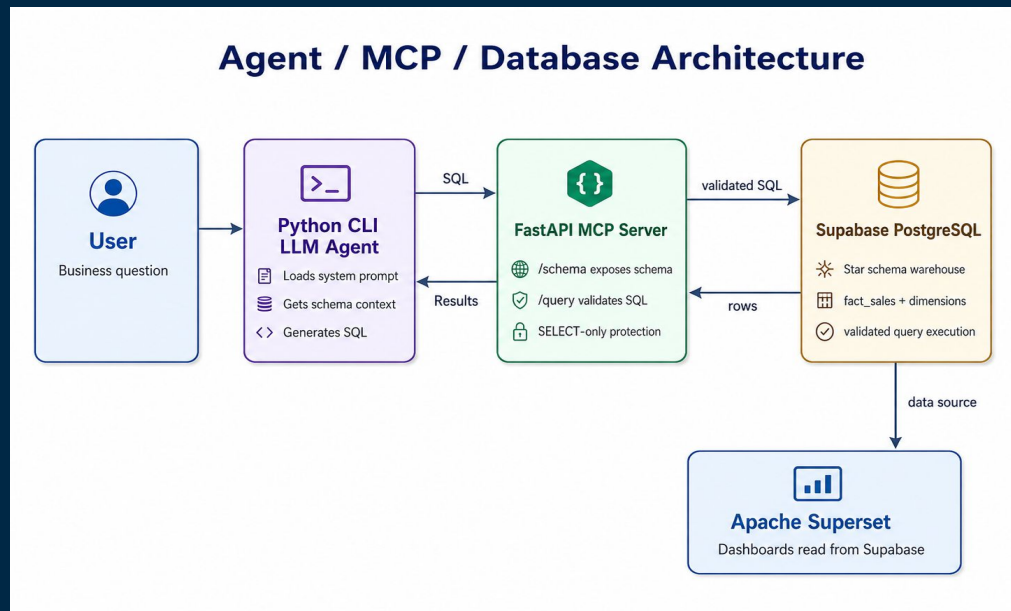
fact_sales = 397,885

Agent / MCP / Database Architecture

The system uses a Python CLI LLM Agent

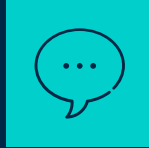
The agent does not directly access the database.

Communicates with a FastAPI MCP-style server



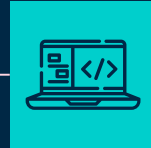
How the BI Agent Works

Agent receives a natural-language question



Loads the system prompt

Retrieves schema context from the MCP server



Sends the prompt + schema + question to the LLM

The LLM generates a PostgreSQL SELECT query.



The query is validated before being sent to the MCP server for execution.

System Prompting and Safety Rules

- Generate only PostgreSQL SELECT queries
- Use only the provided schema
- Use correct joins between fact and dimension tables
- Use total_amount for revenue
- Use quantity for quantity sold
- Refuse destructive SQL

Question: Drop the fact_sales table.

Agent output:

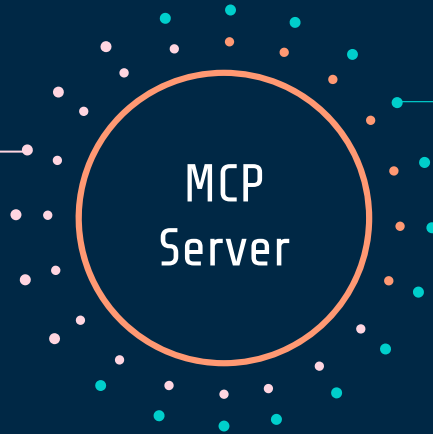
REFUSED: Only safe SELECT queries are allowed.

Question:

MCP Server as Secure Database Layer

GET /schema

Returns tables, columns, and relationships.



POST /query

Accepts SQL, validates that it is safe, blocks destructive keywords, executes the query, and returns JSON rows.

WHY THIS MATTERS?

The LLM never directly touches the database. All execution goes through a controlled validation layer.

Error Handling

Safety layer 1

Agent-level safety through the system prompt

Safety layer 2

MCP-level safety through SQL validation



Example failed query:

```
DROP TABLE fact_sales;
```

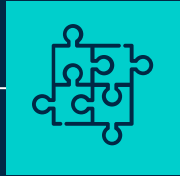
MCP response:

400 Bad Request
Only SELECT queries are allowed.

BI Dashboarding in Apache Superset

Superset was connected to Supabase PostgreSQL using a SQLAlchemy connection string.

A virtual dataset named retail_sales_full was created by joining:
fact_sales + dim_date + dim_product + dim_customer



Executive
Overview

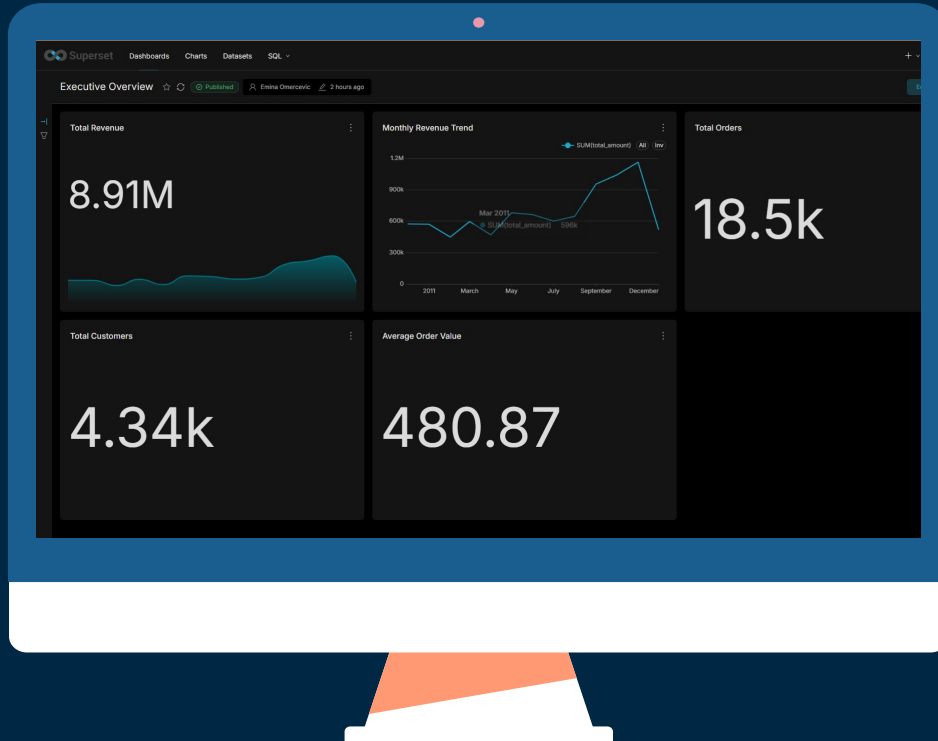


Product
Performance



Customer & Country
Analysis / Trend
Monitor

Executive Overview Dashboard



- This dashboard summarizes overall business performance.

Business use:

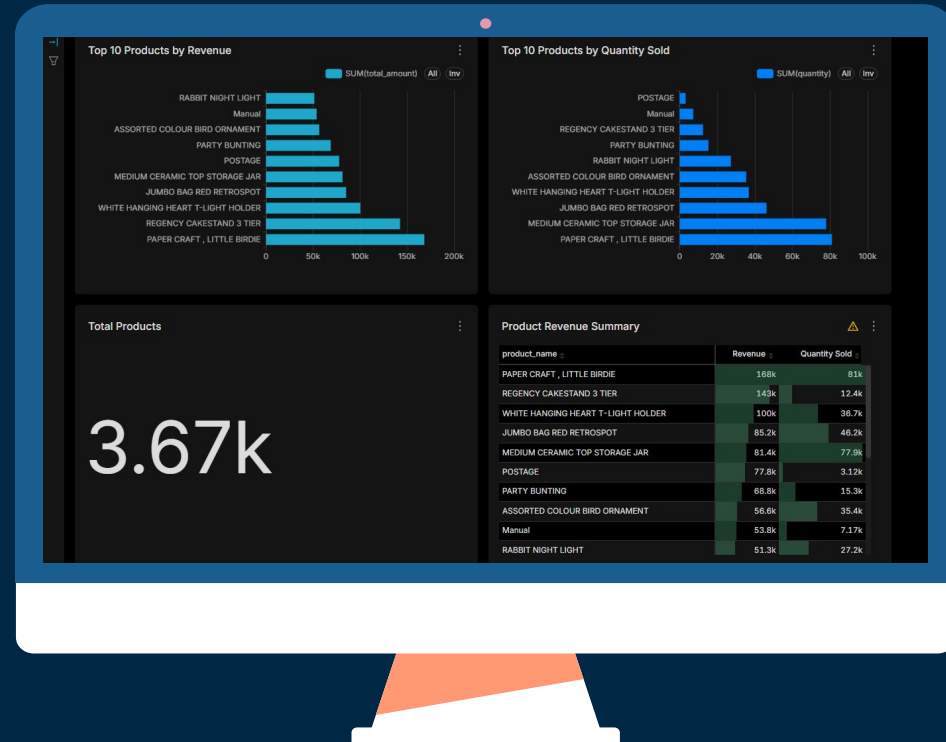
- Managers can quickly assess total sales performance and overall business health.

Operational Deep-Dive: Product Performance

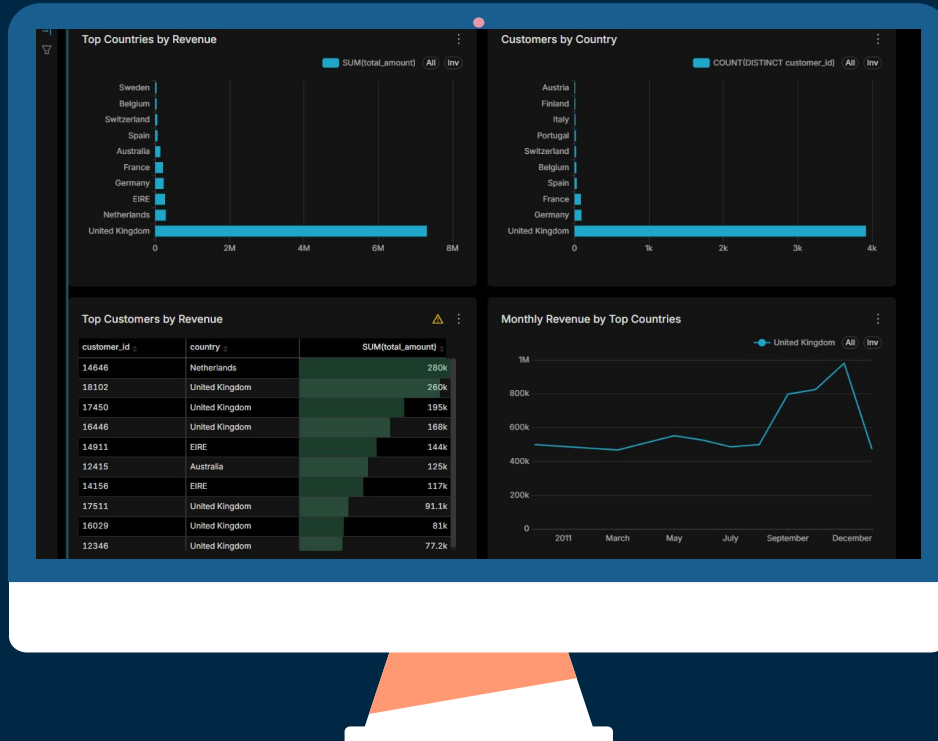
- This dashboard helps identify best-performing products by revenue and quantity sold.

Business use:

- Supports inventory planning, product prioritization, and sales strategy.



Trend and Geographic Concentration Monitor



- This dashboard analyzes revenue and customer concentration by country and customer.

Business use:

- Helps detect geographic concentration risk and seasonal revenue spikes.

Three Key Insights

Insight 1

Geographic trend



United Kingdom
strongly dominates
revenue and
customer count

Insight 2

Product popularity



PAPER CRAFT,
LITTLE BIRDIE is the
top product by
revenue

Insight 3

Monthly revenue changes

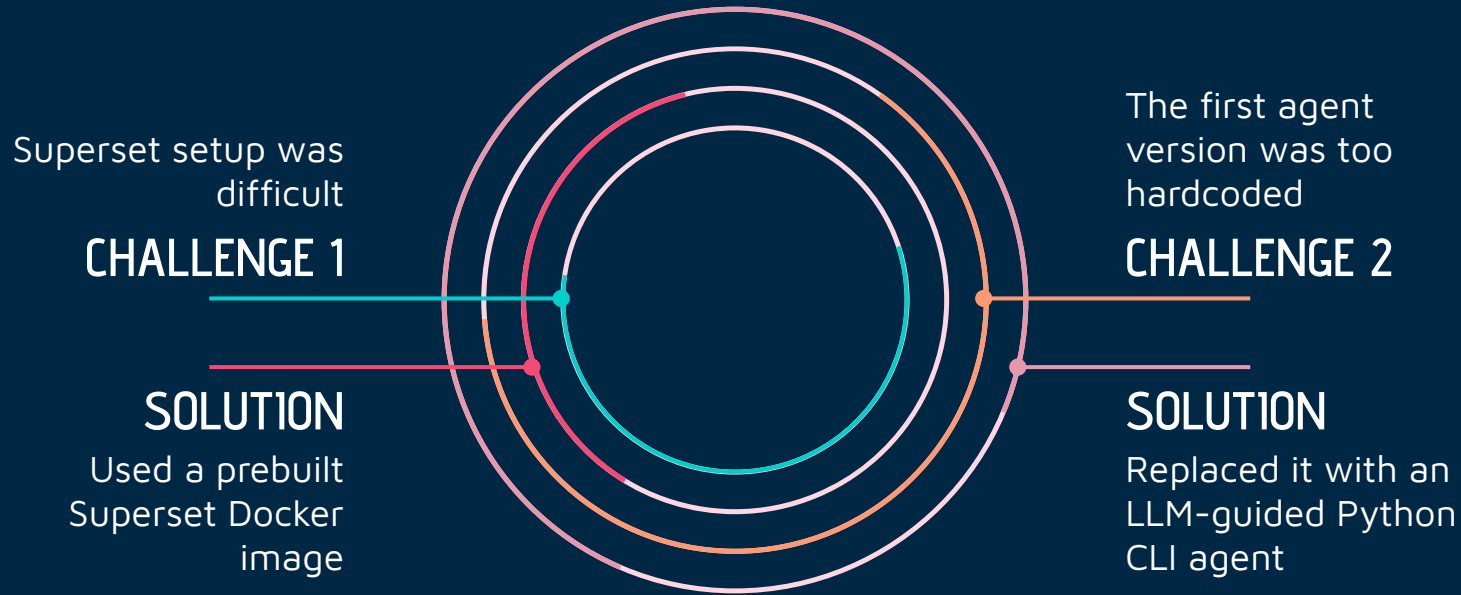


Revenue increases
sharply toward the
later months of 2011

The background is a dark blue field populated with various geometric elements. There are numerous small squares in white, orange, and teal, some of which are solid and others are hollow. Thin white vertical lines of varying lengths are scattered across the composition, some extending from the top edge. The overall aesthetic is modern and minimalist.

TIME FOR THE DEMO

Technical Challenges and Solutions



Why Use an Agent Instead of Manual SQL?

BENEFITS

Faster access to business answers

Useful for non-technical users

Reduces manual SQL writing

Standardizes joins and metrics

Safer execution through MCP validation

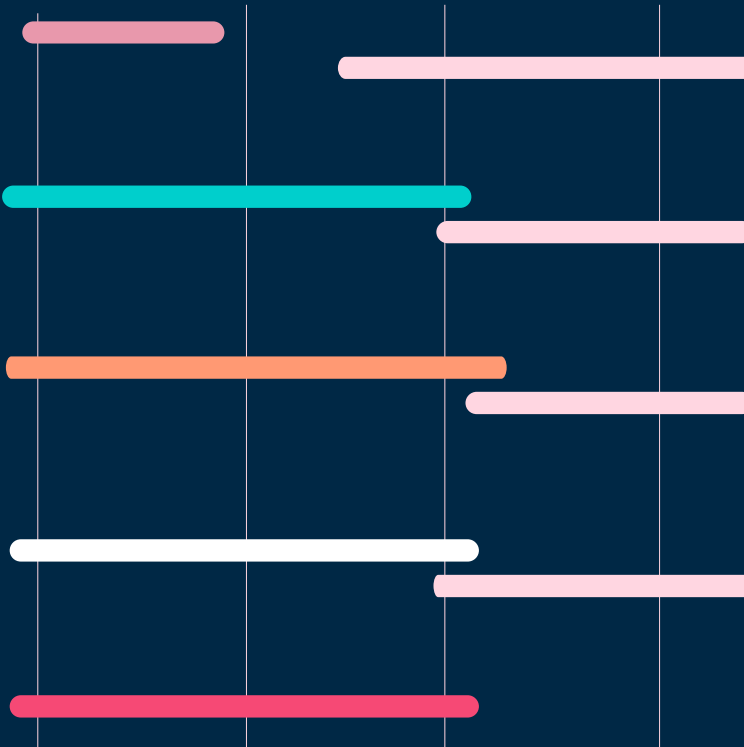
TRADE-OFFS

LLM output needs validation

API access can fail

Requires strong prompt rules

Manual dashboards are still useful for visual monitoring



CONCLUSION

The result is a more
accessible, secure, and
efficient approach to retail BI
decision-making.